

Scratch — Variables Intro

Grade 2 Coding • Let's learn how to use variables!

1. What is Coding?

Coding means giving instructions to a computer.

- Coding tells a computer what to do
- It follows your steps one by one
- Scratch is a fun way to learn coding!

2. Meet Scratch

Scratch is a colorful coding tool made just for kids!

- Drag and snap blocks together
- Make games, stories, and animations
- No typing needed — just click & drag!

3. What is a Variable?

A variable is like a little box where we store information!

Examples of variables:

- Score in a game
- Number of lives
- Player name

The box has a label (name) and something inside (value).

Score:	10
Lives:	3
Name:	Aarav

4. Why Do We Use Variables?

Variables help our program remember!

- To keep track of scores or counts
- To remember things while playing
- To change things as the game goes on

Without variables: The computer forgets everything!

With variables: The score goes up when you win!

5. How to Create a Variable in Scratch

Follow these steps:

- 1 **Step 1:** Click the 'Variables' category (orange blocks)
- 2 **Step 2:** Click 'Make a Variable'
- 3 **Step 3:** Type a name for your variable (like 'score')
- 4 **Step 4:** Click OK

Your new variable appears on the stage!

6. Using Variables in Scratch

Here are the main variable blocks:

set score to 0	gives the variable a starting value
change score by 1	adds 1 to the score
show variable	displays it on the stage
hide variable	hides it from the stage

7. Example: Score Counter

Let's make a score counter game! How it works:

- Set score to 0 at the start
- When the sprite is clicked, change score by 1
- The score shows on the stage!

Try clicking the sprite many times and watch the score go up!

8. Example: Lives Tracker

Now let's track lives in a game! How it works:

- Set lives to 3 at the start
- When something bad happens, change lives by -1
- If lives = 0, game over!

This way the computer remembers how many chances you have left.

9. Let's Try It!

Your turn! Make your own variable. Think about:

- What game will you make?

- What will your variable track? (score, lives, time, coins?)
- What will you name it?

Open Scratch and try creating a variable right now!

10. Recap: What Did We Learn?

Great job! Here's what we learned:

- A variable is a box that stores information
- Variables help us remember and change things in our programs
- We can create variables in Scratch using the orange blocks
- We can set, change, show, and hide variables!

11. Quiz Time!

Question 1: What is a variable?

A) A box to store information B) A type of game C) A color in Scratch

Question 2: Which of these can be a variable?

A) Score B) Player name C) Both A and B

Answers

Question 1 Answer: A) A box to store information

A variable holds data like scores, names, or lives.

Question 2 Answer: C) Both A and B

Score AND player name can both be stored in variables!

12. Next Steps

Keep exploring and practising!

- Try adding variables in your own Scratch projects
- Make a game with a score counter
- Ask a teacher or parent for help
- Practice makes you a better coder!

You're doing amazing — keep going!

Thank you for learning about variables with Scratch! Keep coding and have fun!